CME 250:
Introduction to Machine Learning

Lecture 7:
Unsupervised Learning

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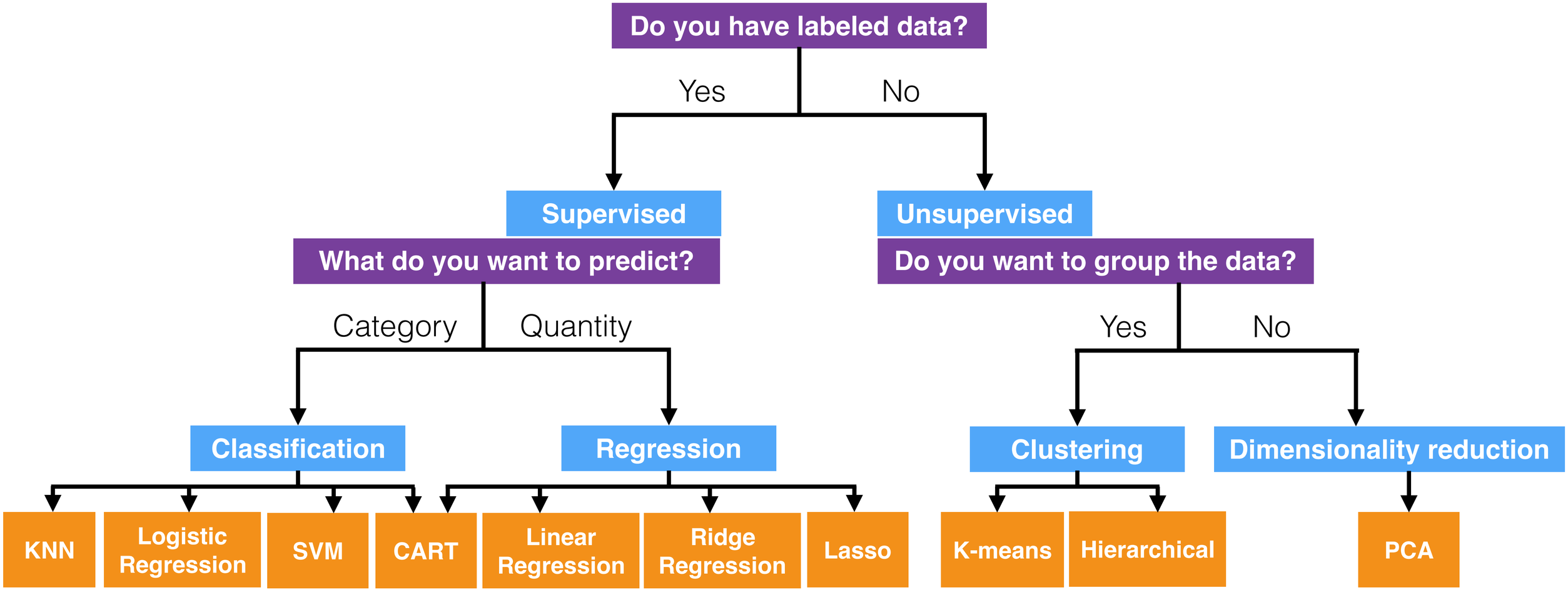
Agenda

Slides are online at
cme250.stanford.edu

- Clustering methods
  - K-means clustering
  - Hierarchical clustering
- Dimensionality reduction
  - PCA

Machine Learning Methods



Unsupervised Learning

Recall: A set of statistical tools for data that only has features/input available, but no response.

In other words, we have X's but no labels y.

Goal: Discover interesting patterns/properties of the data.

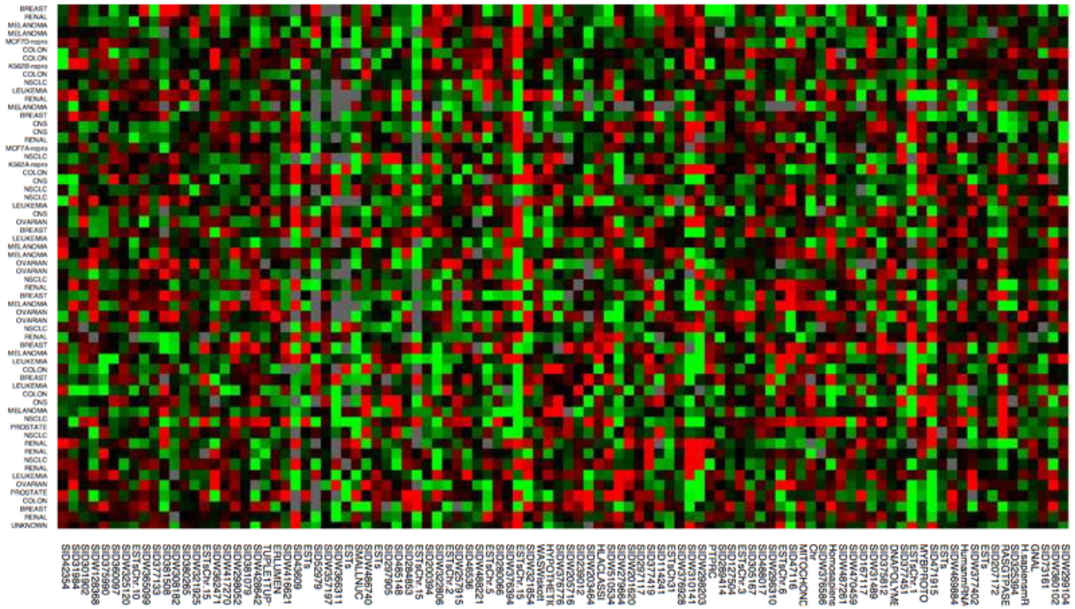
- E.g. for visualizing or interpreting high-dimensional data.

Unsupervised Learning

Example applications:

- Given tissue samples from n patients with breast cancer, identify unknown subtypes of breast cancer.

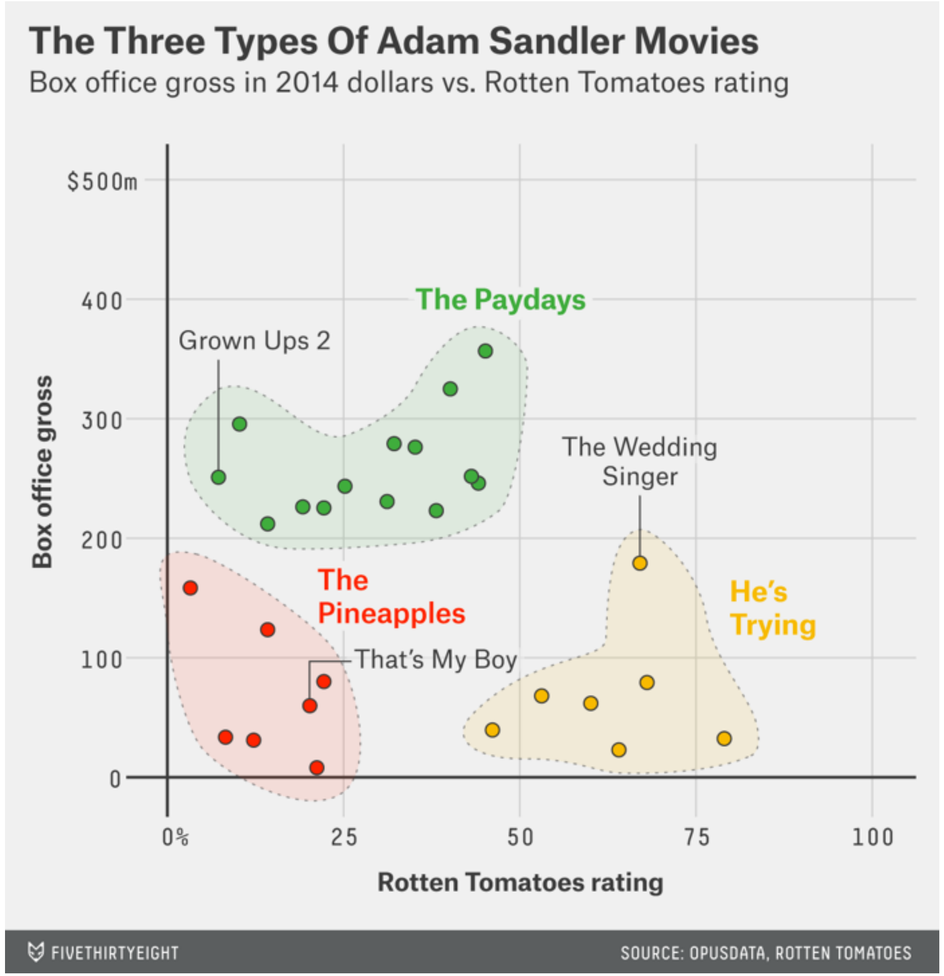
- Gene expression experiments have thousands of variables. Represent the data using a smaller set of features for visualization and interpretation.



Unsupervised Learning

Example applications:

- Document clustering: identify sets of documents about the same topic.
- Given high-dimensional facial images, find a compact representation as inputs for a facial recognition classifier.



Challenges of Unsupervised Learning

Why is unsupervised learning challenging?

- Exploratory data analysis — goal is not always clearly defined
- Difficult to assess performance — “right answer” unknown
- Working with high-dimensional data

Types of Unsupervised Learning

Two approaches:

- **Cluster analysis**
  - For identifying homogenous subgroups of samples
- **Dimensionality reduction**
  - For finding a low-dimensional representation to characterize and visualize the data

Cluster Analysis

Clustering

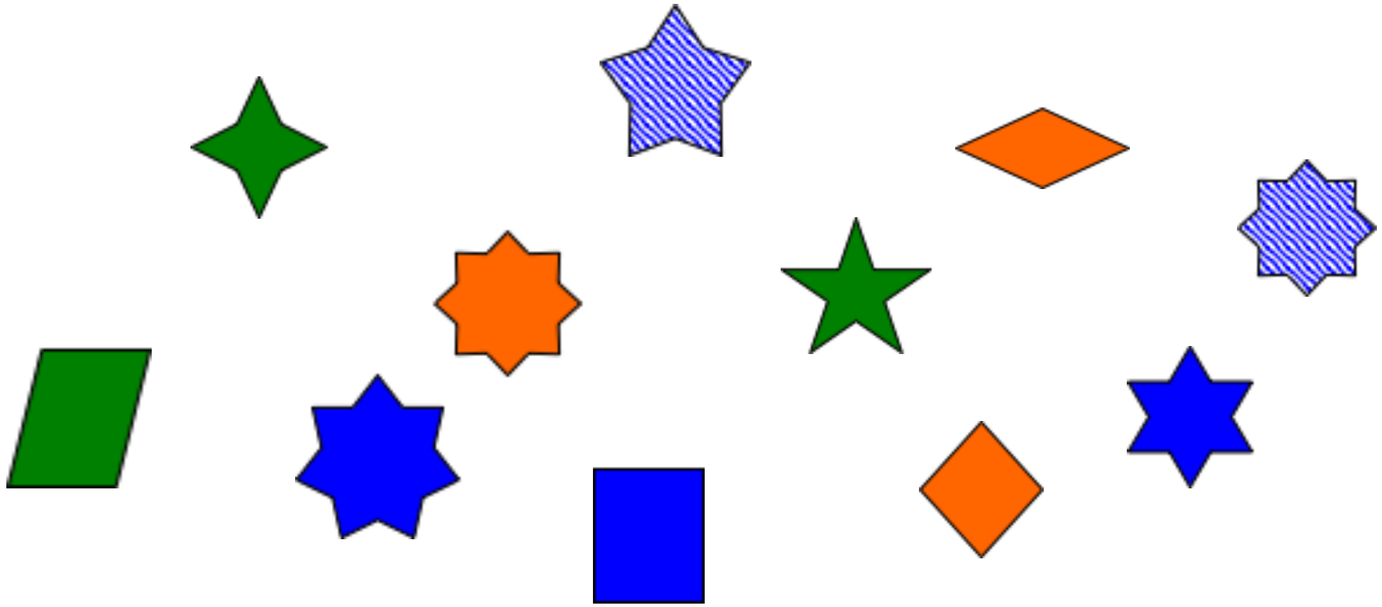
A set of methods for finding subgroups within the dataset.

- Observations should share common characteristics within the same group, but differ across groups.

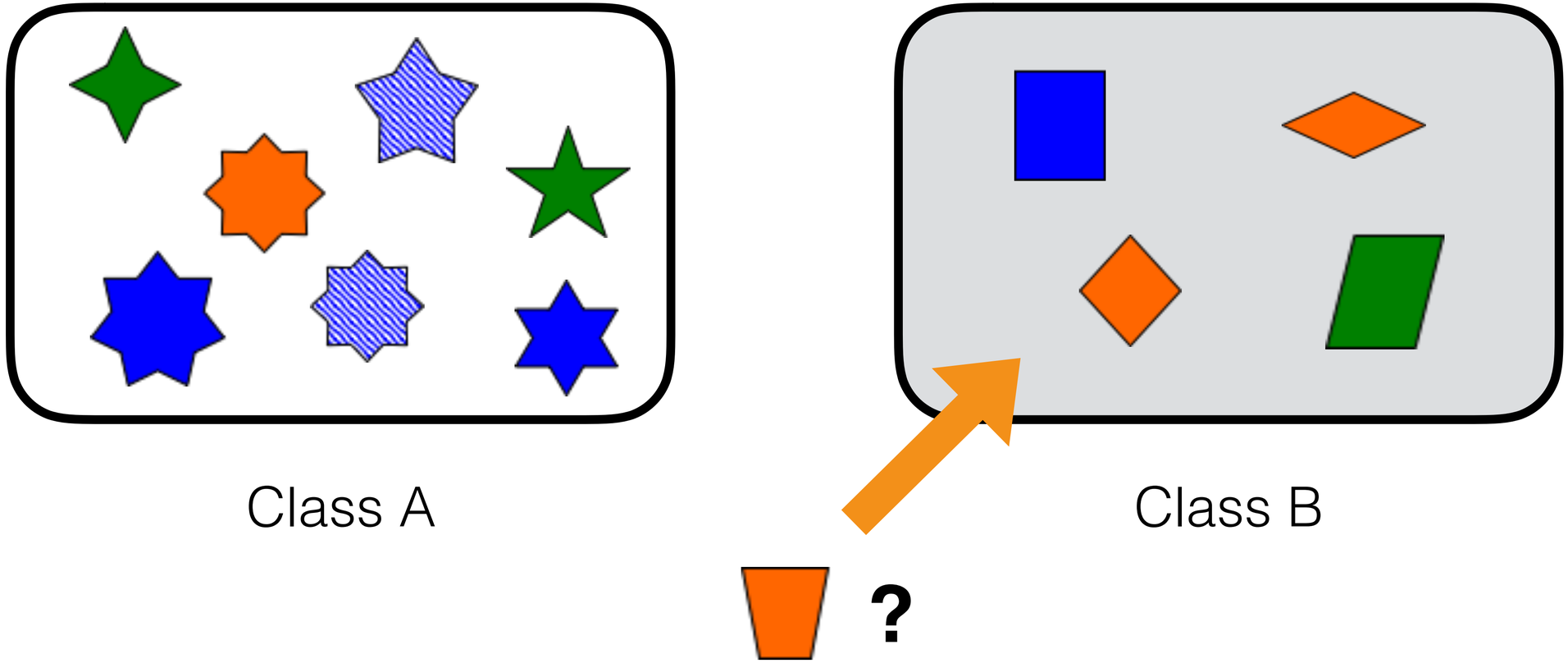
- Groupings are determined from attributes of the data itself — differs from classification.

https://medium.com/square-corner-blog/so-you-have-some-clusters-now-what-abfd297a575b

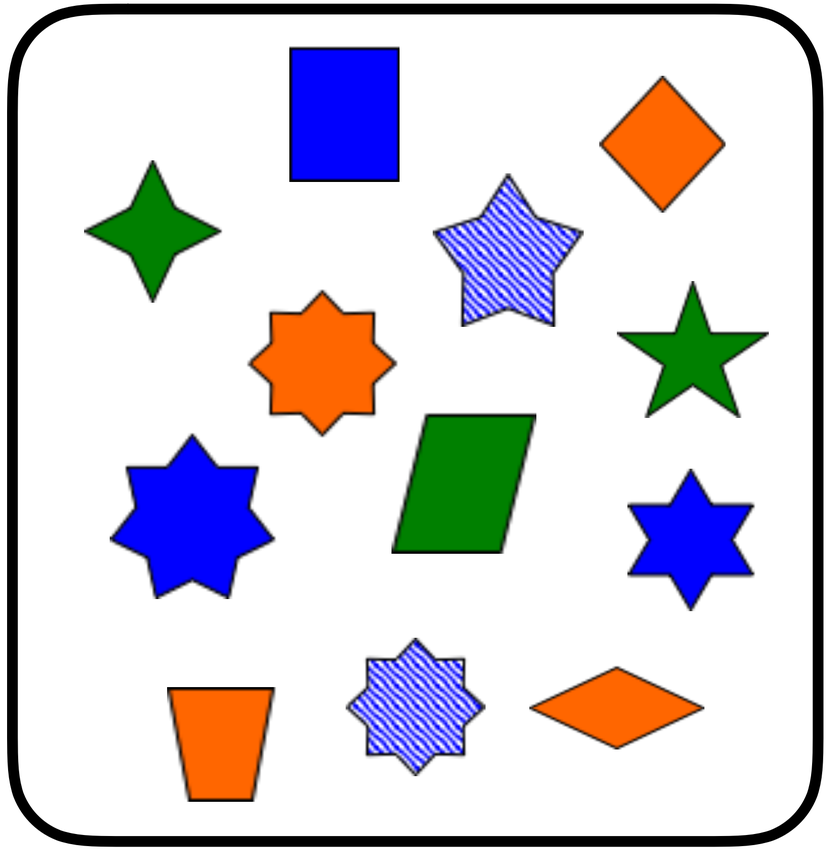
Clustering vs. Classification



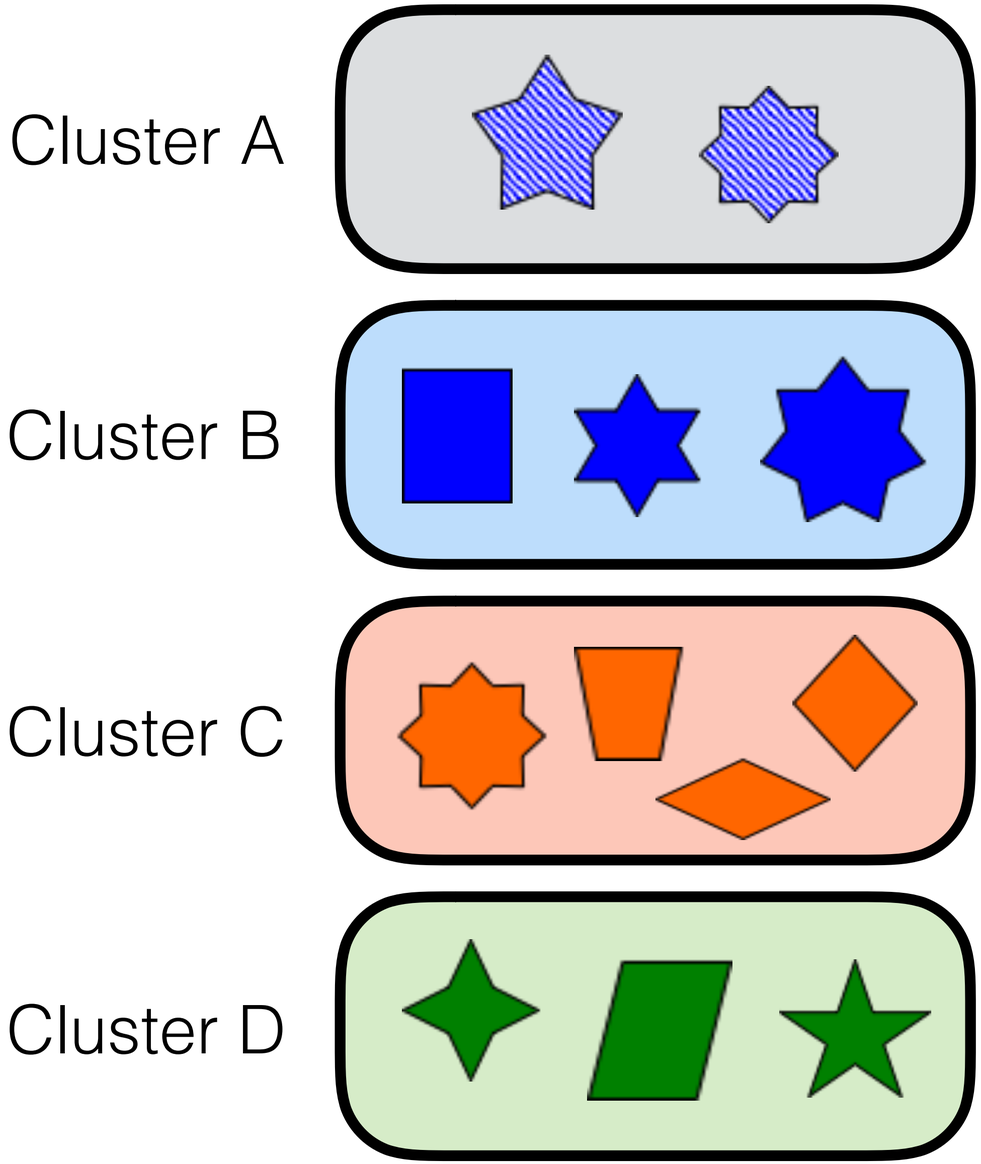
Classification



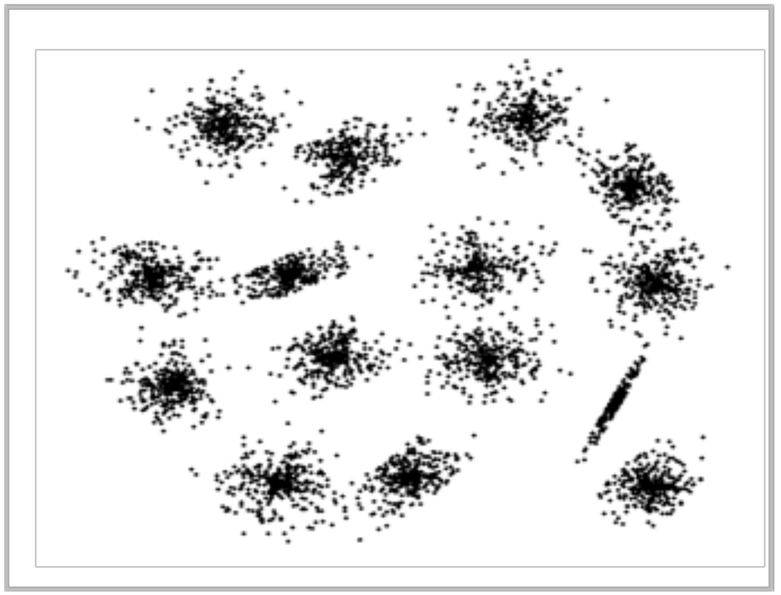
Clustering



Dataset



Clustering



http://cs.joensuu.fi/sipu/datasets/

Types of Clustering

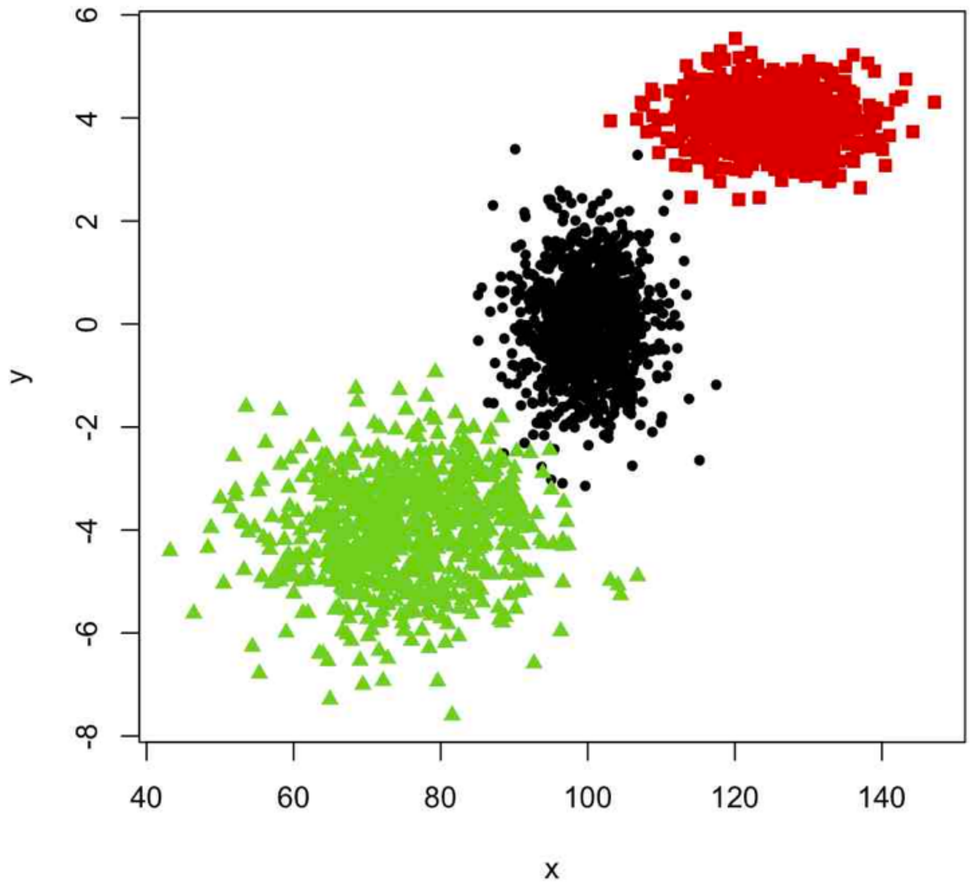
- **Centroid-based clustering**
- **Hierarchical clustering**
- **Model-based clustering**
  - Each cluster is represented by a parametric distribution
  - Dataset is a mixture of distributions
- **Hard vs. soft/fuzzy clustering**
  - Hard: observations divided into distinct clusters
  - Soft: observations may belong to more than one cluster

K-means Clustering

Groups data into K clusters that satisfy two properties.

1. Each observation belongs to at least one of the K clusters.

2. Clusters are non-overlapping. No observation belongs to more than one cluster.



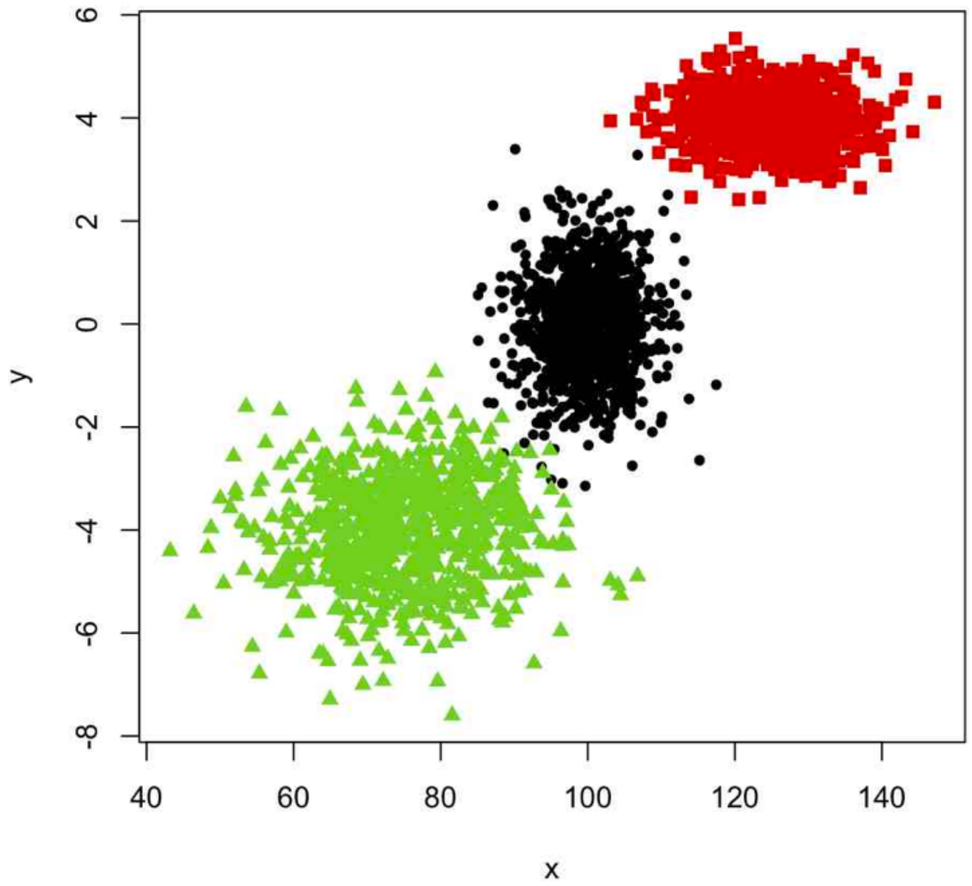
K-means Clustering

A good clustering is one for which the *within-cluster variation* is as small as possible.

Denote each cluster by C_k, and let $W(C_k)$ be a measure of the within-cluster variation.

K-means aims to solve

$$\underset{C_1,\ldots,C_K}{\text{minimize}} \left\{ \sum_{k=1}^{K} W(C_k) \right\}$$



K-means Clustering

How to measure within-cluster variation?

The most common choice is squared Euclidean distance.

$$W(C_k) = \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2$$

Which means overall we solve

$$\underset{C_1,\ldots,C_K}{\text{minimize}} \left\{ \sum_{k=1}^{K} \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2 \right\}$$

K-means Clustering

It turns out that this optimization problem is difficult to solve, as it is discrete and there are nearly K^n ways to split n samples into K clusters.

In practice, use an iterative algorithm that finds a local minimum to this optimization.

K-means Clustering Algorithm

1. Initialize each observation to a cluster by randomly assigning a cluster, from 1 to K, to each observation.

2. Iterate until the cluster assignments stop changing:

   a. For each of the K clusters, compute the cluster centroid. The k-th cluster centroid is the vector of the p feature means for the observations in the k-th cluster.

   b. Assign each observation to the cluster whose centroid is closest (using Euclidean distance as the metric).

K-means Clustering Iterations

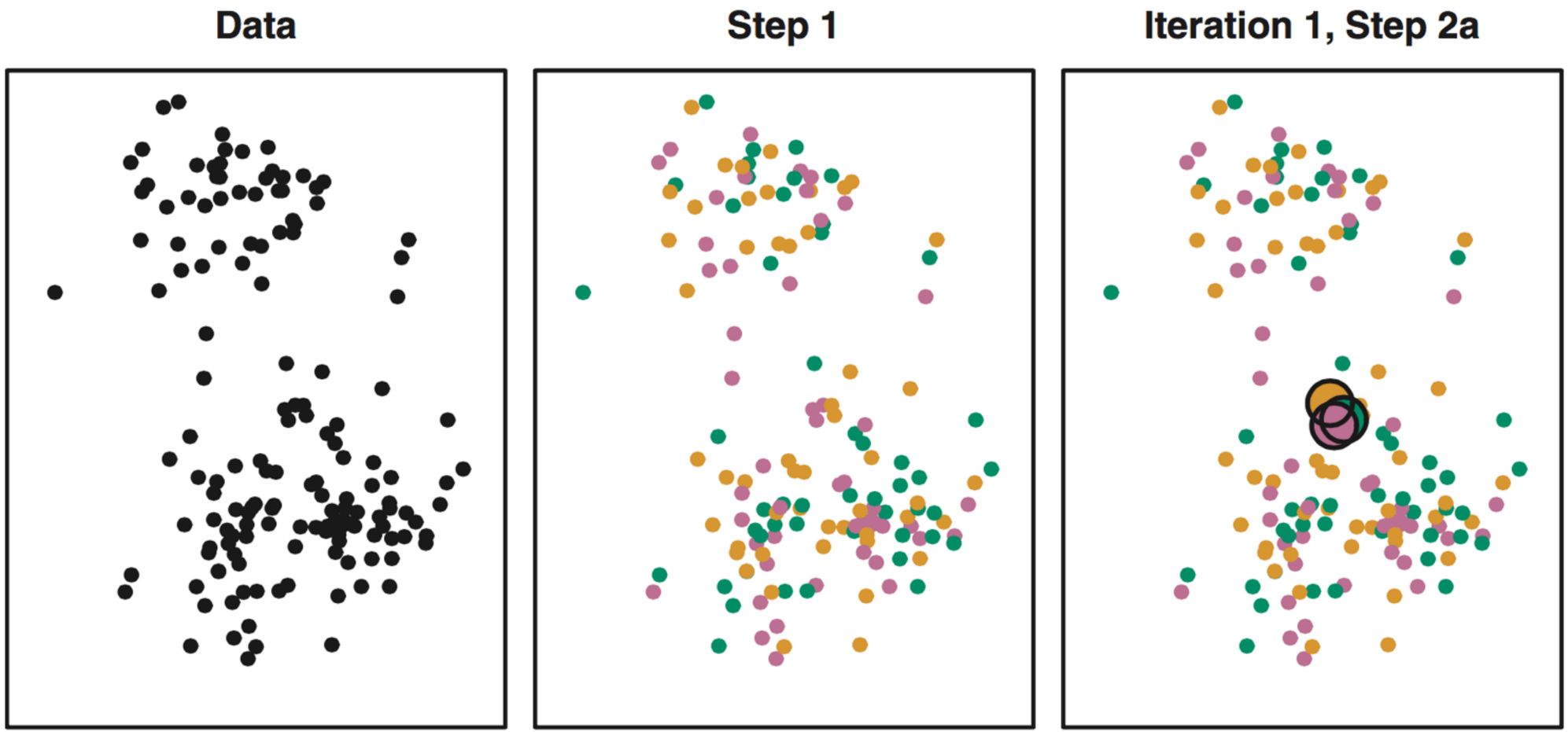


FIGURE 10.6, ISL (8th printing 2017)

K-means Clustering Iterations

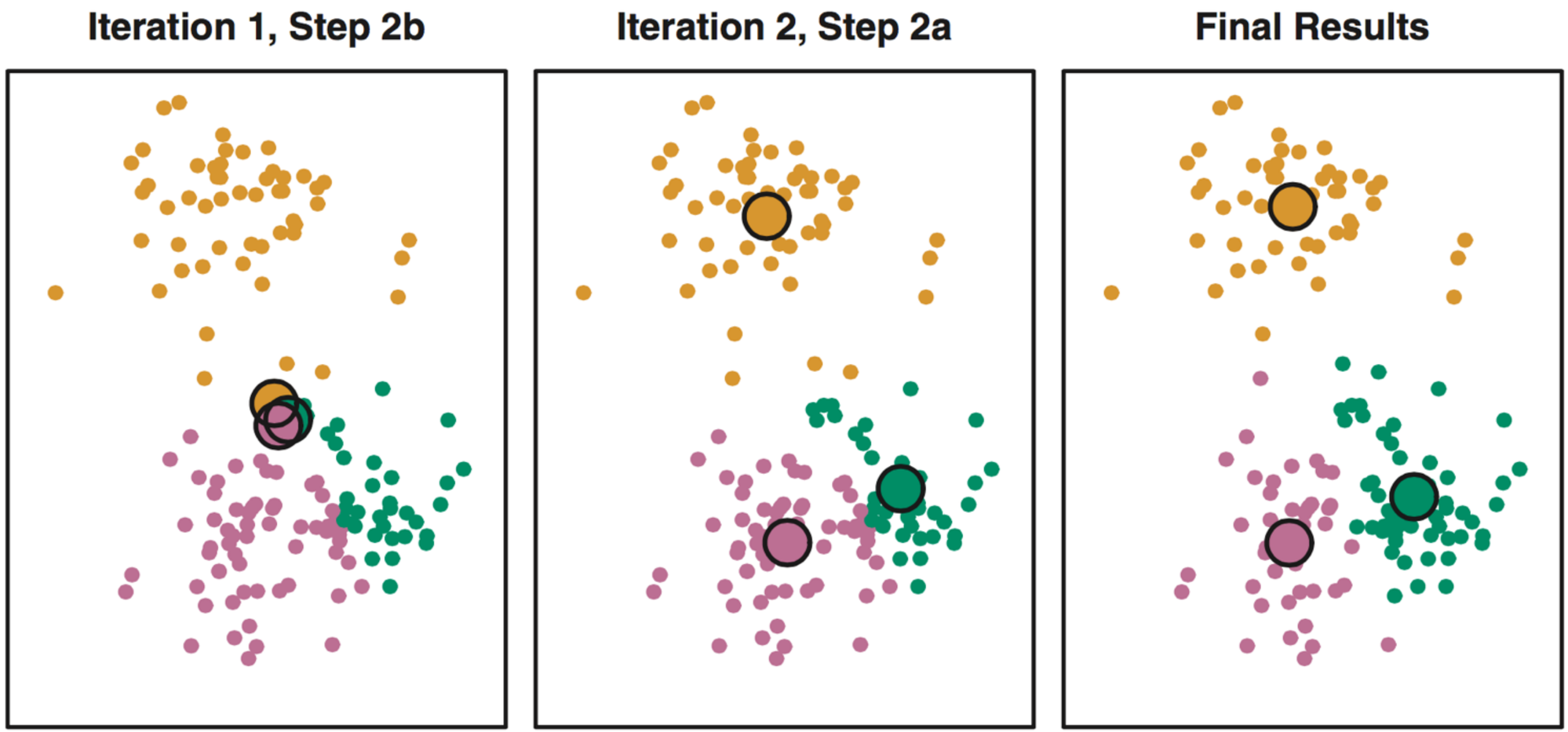
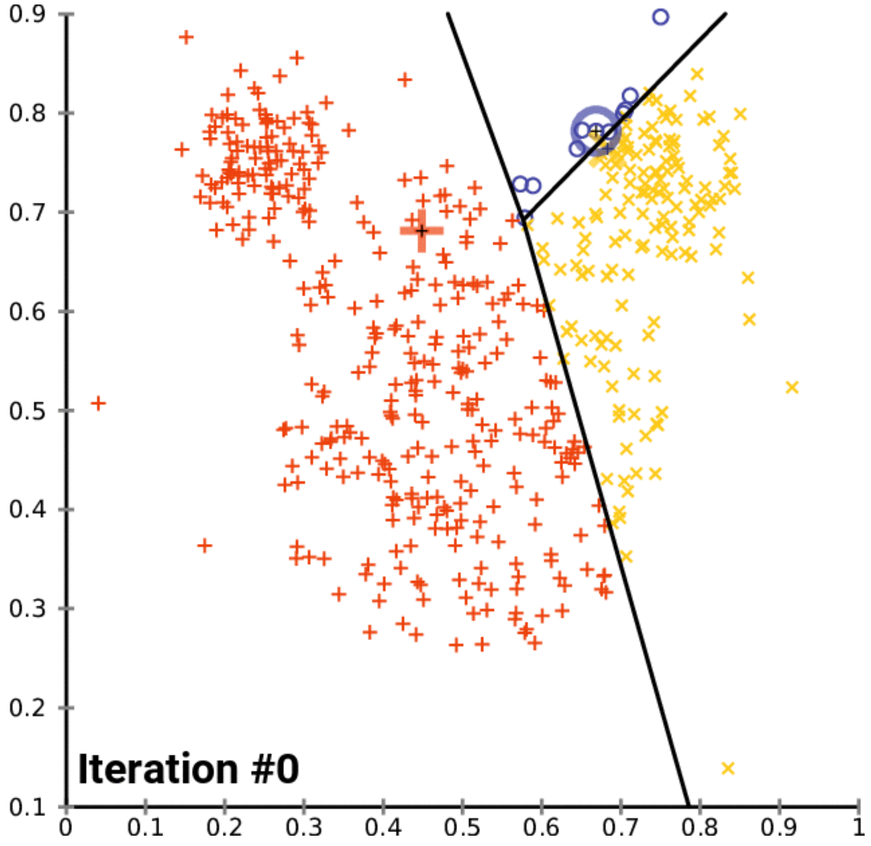


FIGURE 10.6, ISL (8th printing 2017)

K-means Clustering Animation



K-means Clustering Properties

It can be shown that the value of the objective function will never increase at each iteration of k-means.

Since the algorithm finds local minima, however, it will result in different clusters with different initializations.

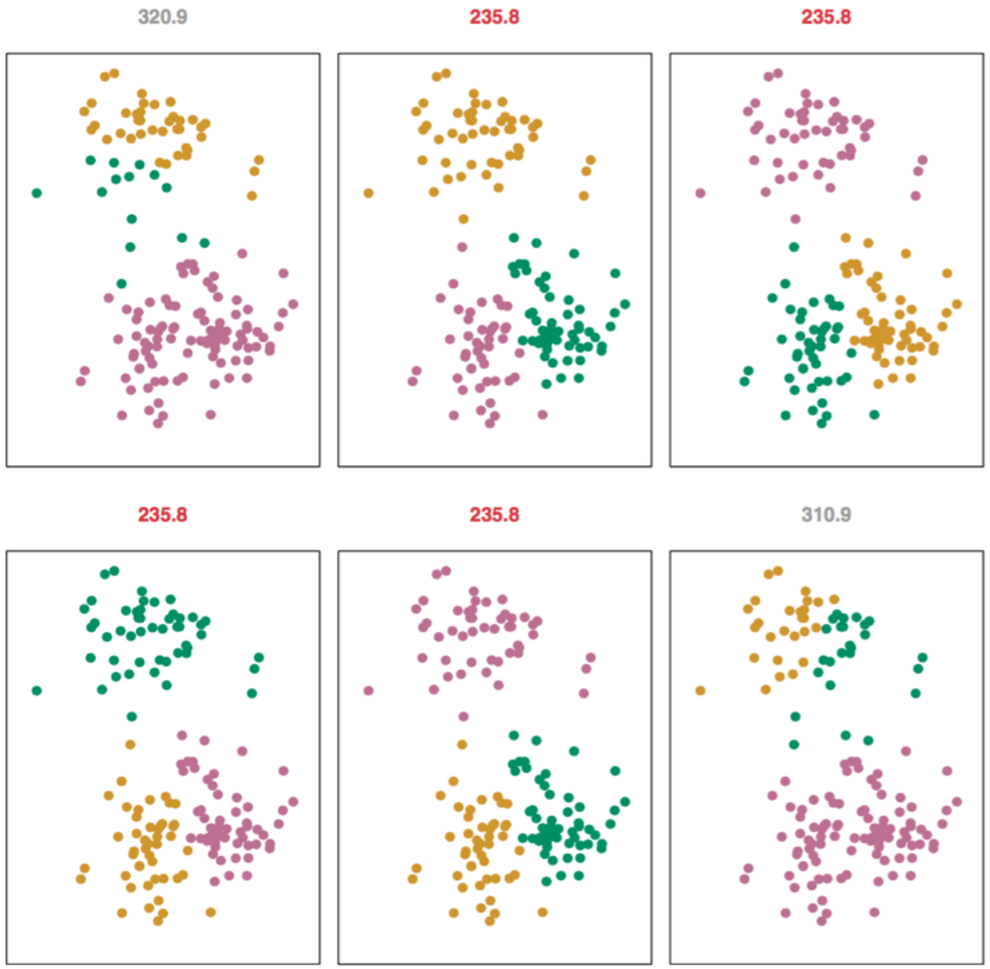


FIGURE 10.7, ISL (8th printing 2017)

K-means Pros and Cons

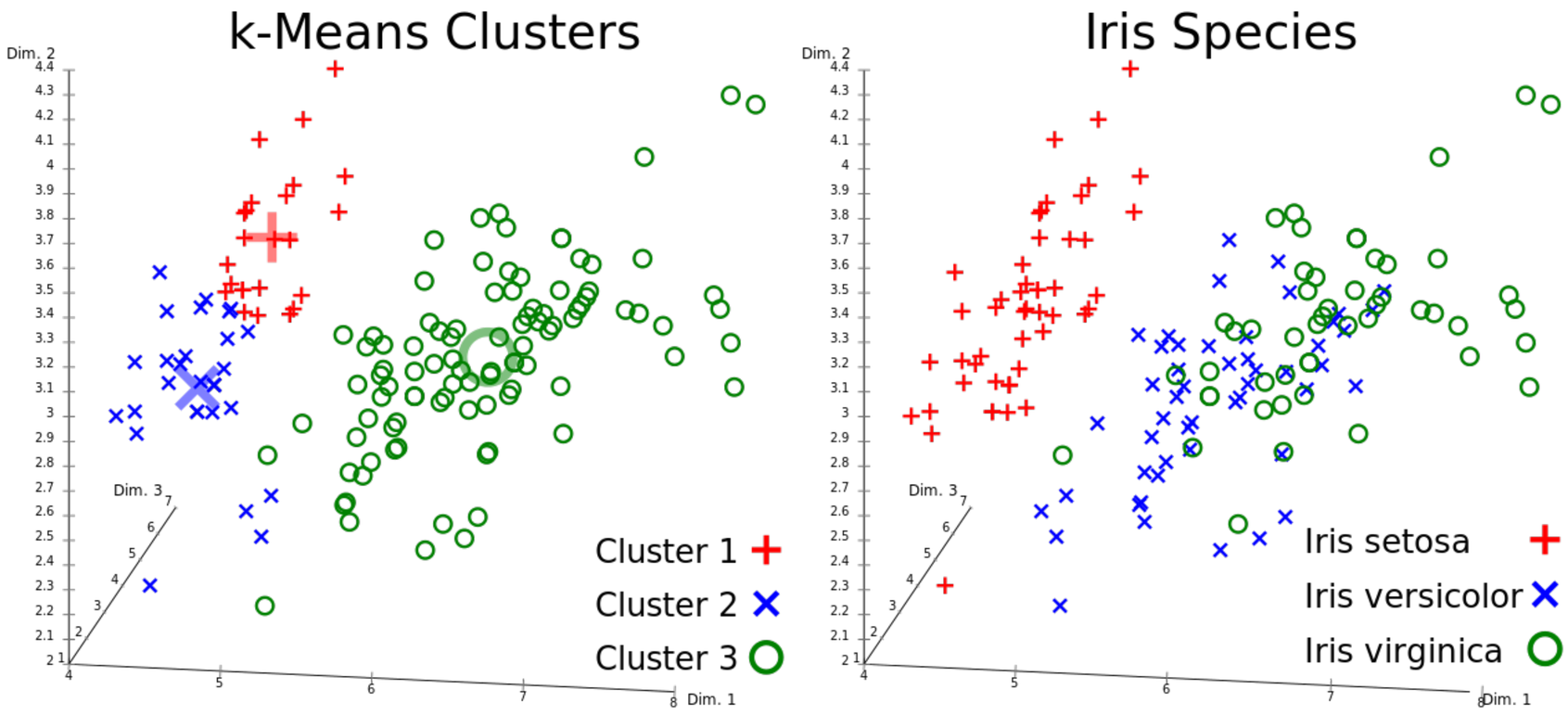
Pros:

- Easy to implement and understand

Cons:

- Not robust to data perturbations and different initializations
- Treats each feature equally, not robust to noise features or different scales of features — looks for in spherical clusters in feature space
- Need to define K before running algorithm

Another K-means Example



Dimensionality Reduction

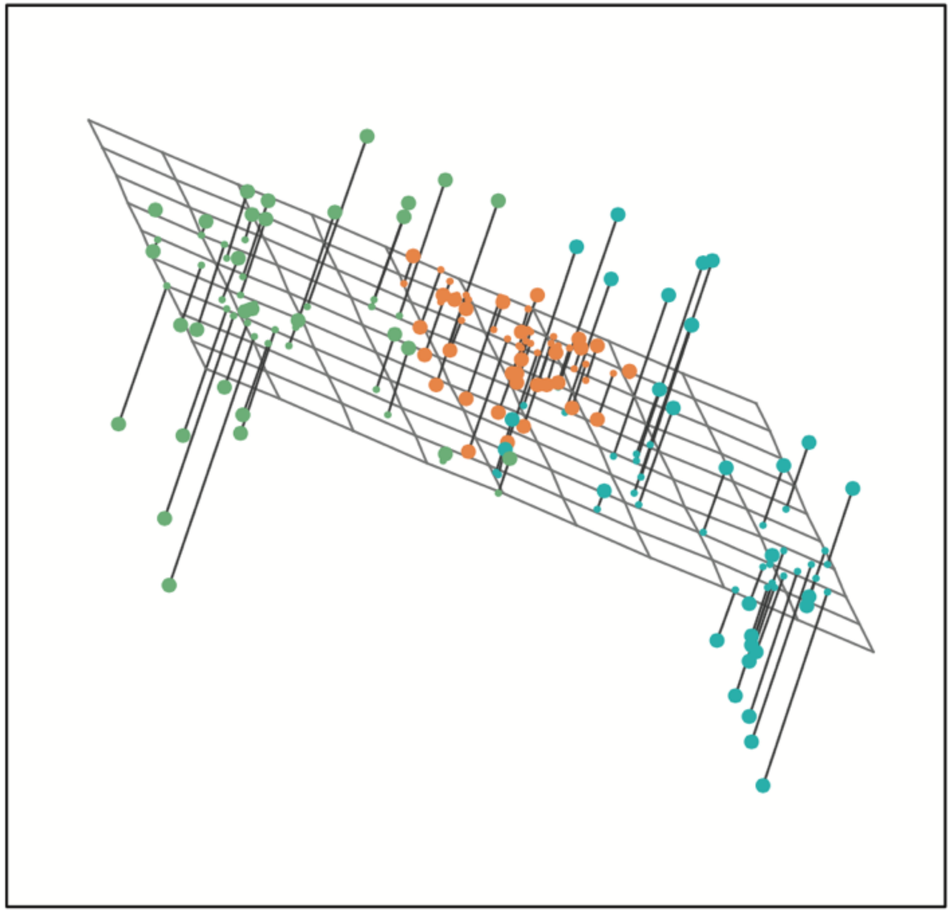
Dimensionality Reduction

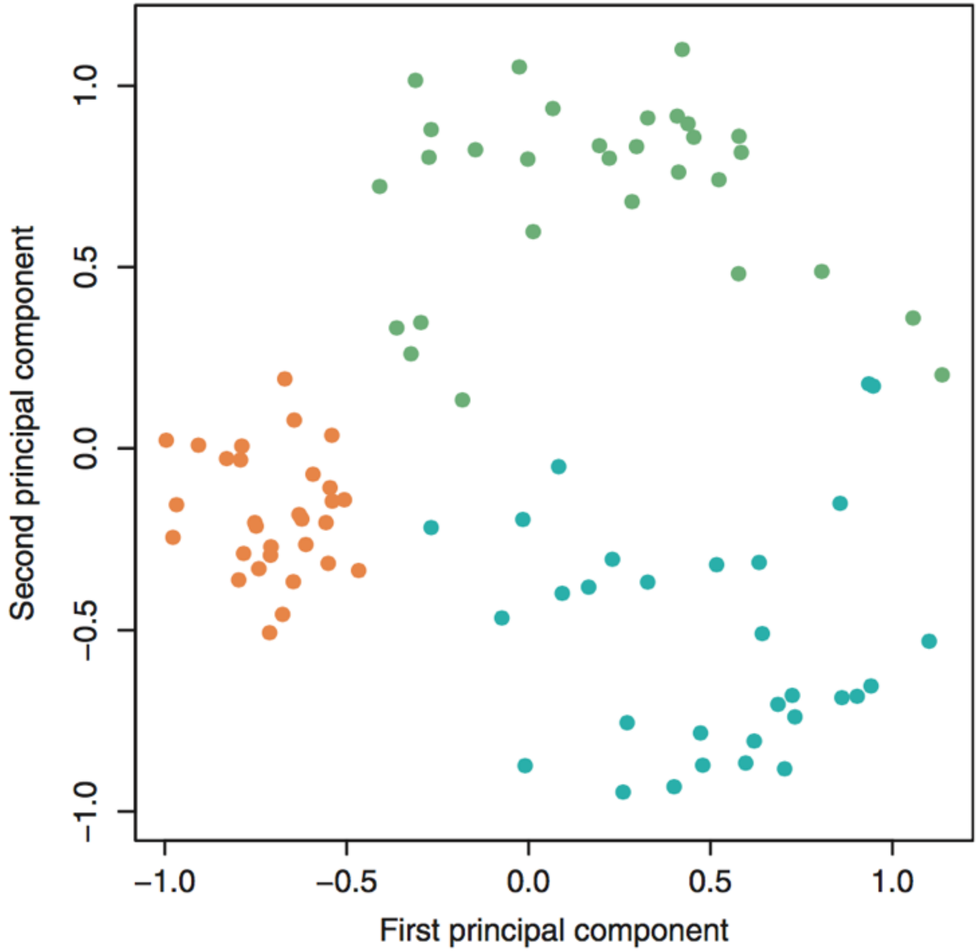
Recall the curse of dimensionality when working in high dimensions.

Dimensionality reduction is the process of reducing the number of features under consideration.

We already saw some examples of this in the lasso and forward/backward selection algorithms. These methods reduce dimensionality by selecting a subset of features. However, they do so using supervision — knowing a response y that is of interest.

Dimensionality Reduction





Principal Component Analysis

Look for a low-dimensional representation of the dataset that contains as much *variation* in the dataset as possible.

E.g. for plotting our data and gaining intuition, if we can obtain a 2D representation of the data, then we can plot the observations in this low-dimensional space.

Note that you want to *center* the data and make the scales of features comparable before performing PCA. E.g. if one feature is in kilometers and another in meters, the one in kilometers may appear to have lower variance when in fact this is due to scaling.

Principal Components

The first principal component of a set of features $X_1, X_2, \ldots, X_p$ is the normalized linear combination of the features

$$Z_1 = \phi_{11} X_1 + \phi_{21} X_2 + \ldots + \phi_{p1} X_p$$

that has the largest variance. "Normalized" refers to $\sum_{j=1}^{p} \phi_{j1}^2 = 1$.

We refer to $\phi_{11}, \ldots, \phi_{p1}$ as the *loadings* of the first principal component.

The loadings make up the first principal component vector.

$$\phi_1 = (\phi_{11} \;\; \phi_{21} \;\; \ldots \;\; \phi_{p1})^T$$

Principal Components

The first principal component loading vector solves the optimization problem

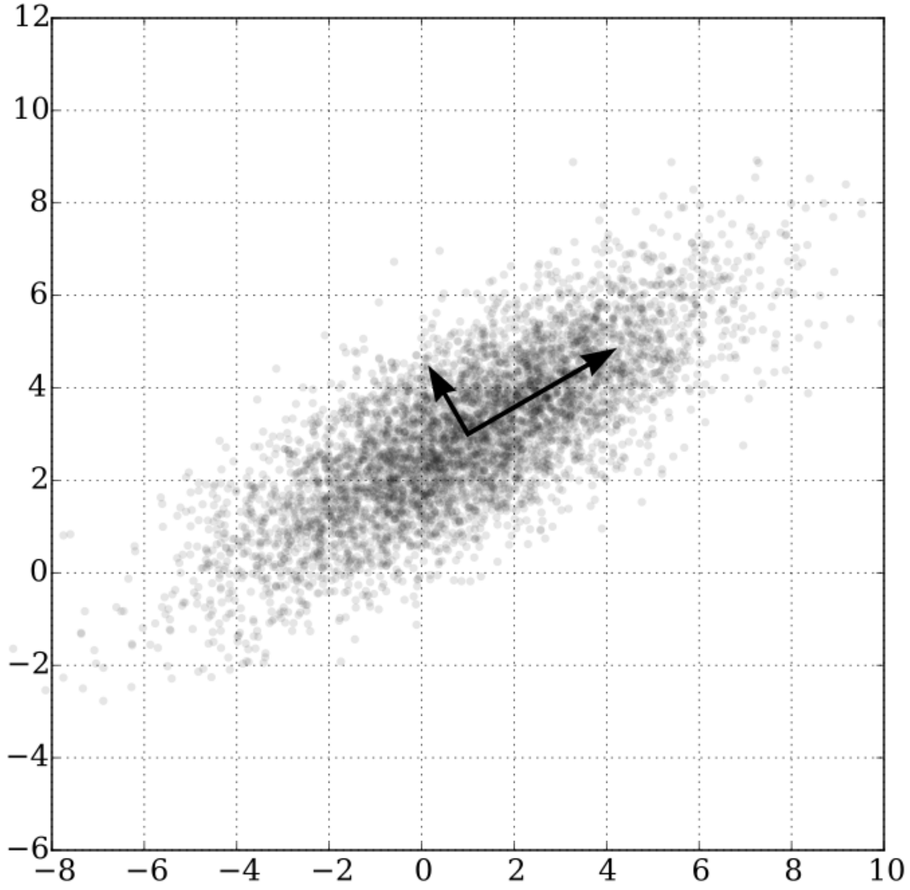
$$\underset{\phi_{11},\ldots,\phi_{p1}}{\text{maximize}} \left\{ \frac{1}{n} \sum_{i=1}^{n} \left(\sum_{j=1}^{p} \phi_{j1} x_{ij} \right)^2 \right\} \text{ subject to } \sum_{j=1}^{p} \phi_{j1}^2 = 1$$

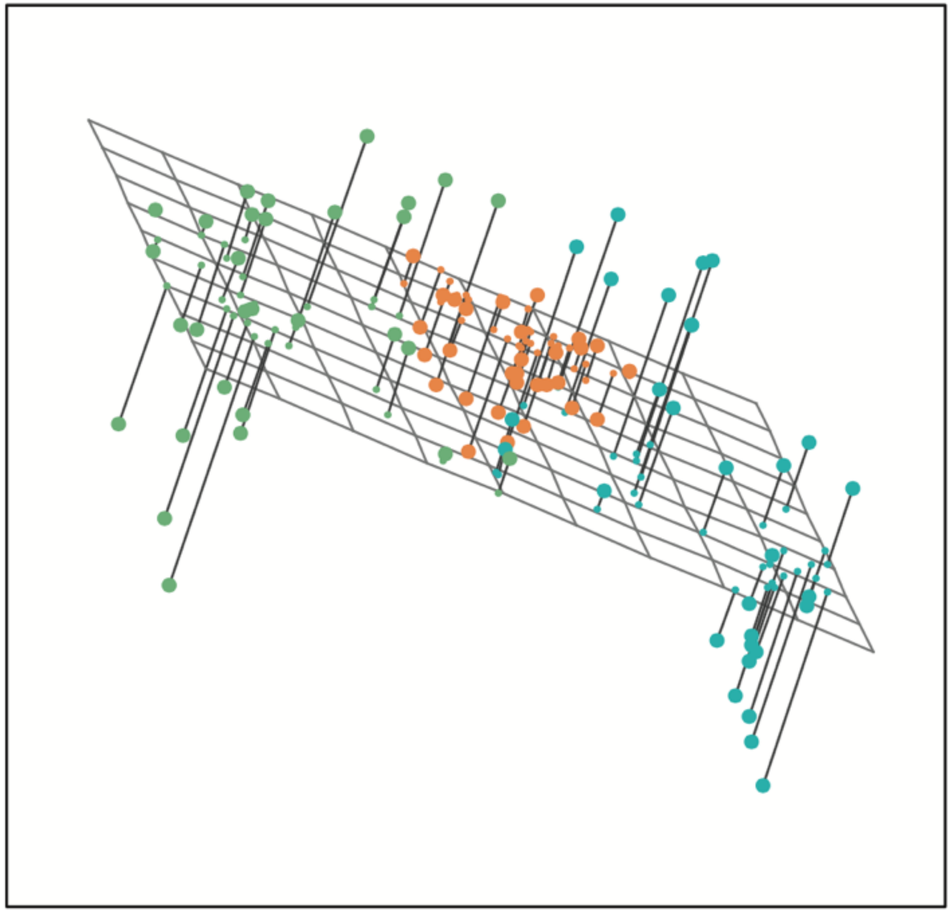
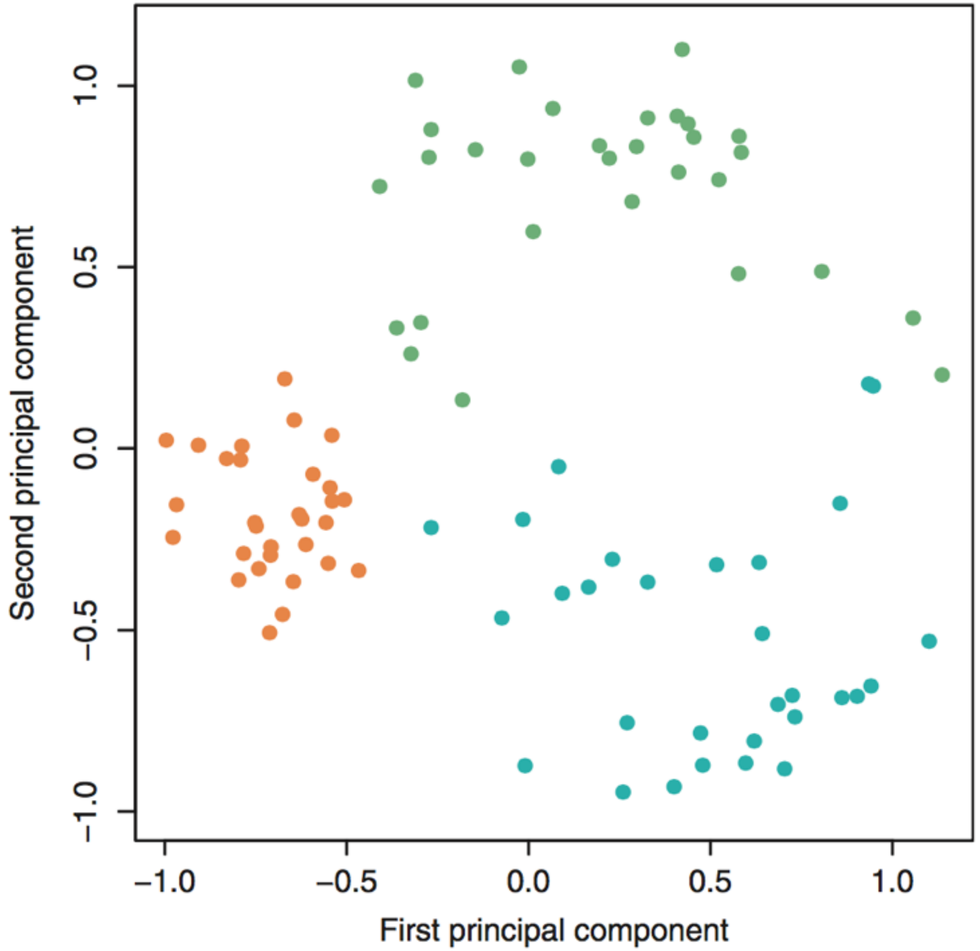
Principal Components

The second principal component Z_2 is the linear combination of features that has maximal variance out of all linear combinations that are are uncorrelated with Z_1.

Constraining Z_2 to be uncorrelated with Z_1 is equivalent to constraining the direction of ϕ_2 to be orthogonal to ϕ_1 .

Principal Component Analysis



First two principal axes of this Gaussian dataset.

Principal Component Analysis